LESSON 10.2

Math is Beauty

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use bilateral symmetry to describe structure in nature and in human creations.

WHY THIS MATTERS There are over 17,000 species of butterflies in the world, and most people are fascinated by them. Part of the fascination is mathematical: math creates beautiful designs, many of which are found naturally in plants and animals.

REMEMBER

Math is Beauty

Symmetry, Geometry & Design Look closely at a butterfly's wings, then at how a plant grows its leaves. The same mathematical idea keeps appearing.

1. Line of Symmetry: Divides a figure into two congruent mirror-image halves.
2. Rotational Symmetry: Shape rotates onto itself in less than a full 360° turn.
3. Geometric transformations preserve angle measures and side lengths

Words: symmetric *(simétrico)* · mirror image *(imagen reflejada)* · bilateral symmetry *(simetría bilateral)* · line of symmetry *(eje de simetría)* · balance *(equilibrio)*

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each figure by whether it has a line of symmetry.

Groups: Has a line of symmetry · No line of symmetry

_____ A butterfly with open wings

_____ The letter A

_____ A square

_____ A valentine heart

_____ The letter F

_____ The letter J

_____ The letter R

_____ A flag shape like the letter P

HINT Run the mirror test: can you draw a line so each half is the other's mirror image?

2 **Complete the table. Show how you found each value.**

How many lines of symmetry does each figure have?

Figure	Think about the folds	Lines of symmetry
2 through the sides + 2 through the corners		
through the sides only — corner folds fail		
one through each corner		
one vertical + one horizontal		

HINT Count every fold that maps the figure onto itself — sides AND diagonals.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A rectangular garden is 8 ft by 5 ft. What is its area?

(A) 13 square feet

(B) 26 square feet

(C) 40 square feet

(D) 40 cubic feet

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

A box measures 3 by 4 by 2 units. What is its volume?

(A) 9 cubic units
(B) 12 cubic units
(C) 24 cubic units
(D) 24 square units

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

5 **Circle the letter of the best answer. Show how you know.**

A recipe makes 4 servings using 2 cups of rice. How much rice is in one serving?

(A) 1/2 cup
(B) 2 cups
(C) 4 cups
(D) 8 cups

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Answer in complete sentences.**

The lesson says bilateral symmetry influences human creations 'from art to music to architecture.' Pick ONE of those three and describe how symmetry appears in it.

SENTENCE STARTER In ___ , symmetry appears when ___ .
Designers use it because ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Find something in this room that you think has bilateral symmetry and your partner thinks does not. Where exactly would the line of symmetry go, and what makes the two halves mirror images rather than just similar?

SENTENCE STARTER ___ has a line of symmetry ___ because the two halves are ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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LEARNING TARGET I can use bilateral symmetry to describe structure in nature and in human creations.

WHY THIS MATTERS There are over 17,000 species of butterflies in the world, and most people are fascinated by them. Part of the fascination is mathematical: math creates beautiful designs, many of which are found naturally in plants and animals.

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which statement best shows what this lesson means by 'math is beauty'?

(A) Only things made by mathematicians can be beautiful

(B) Butterflies are the only living things with symmetry

(C) Beauty and math are opposites — one is feeling and one is numbers

(D) Math creates beautiful designs like bilateral symmetry, found in nature and used in human art, music, and architecture

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A chef counts how many of each ingredient are left in the kitchen. This count is called…

(A) area

(B) volume

(C) a portion

(D) inventory

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Your friend insists that only one of two flowers has bilateral symmetry. One flower's petals mirror each other across a line down its center; the other flower does too, just along a different line. A butterfly's two wings look like mirror images of each other across a line down the center of its body. What is this type of symmetry called?

(A) Rotational symmetry

(B) Radial symmetry

(C) Bilateral symmetry

(D) Point symmetry

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Your friend insists that only one of two flowers has bilateral symmetry. One flower's petals mirror each other across a line down its center; the other flower does too, just along a different line. To test whether a flower has bilateral symmetry, what should you do first?

(A) Count the number of petals

(B) Draw a line through the center and compare the two sides

(C) Measure the flower's height

(D) Compare the flower to a different type of flower

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Your friend insists that only one of two flowers has bilateral symmetry. One flower's petals mirror each other across a line down its center; the other flower does too, just along a different line. Your friend says only one of two flowers has bilateral symmetry. You test both by drawing a line through each flower's center and find both sides match on both flowers. What is most likely true?

(A) The friend is right — only one flower truly has symmetry

(B) Both flowers have bilateral symmetry, but the second flower's line of symmetry likely runs in a direction the friend didn't try

(C) Symmetry cannot be tested in flowers

(D) The two flowers must be identical in shape to both be symmetric

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Find something in this room that you think has bilateral symmetry and your partner thinks does not. Where exactly would the line of symmetry go, and what makes the two halves mirror images rather than just similar?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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